



Vidyavardhini's College of Engineering and Technology, Vasai

Department of Computer Science & Engineering (Data Science)

EXPERIMENT ASSESSMENT

ACADEMIC YEAR 2025-26

Course: Introduction to Web Technology

Course code: 2344211

Year: SE SEM: IV

Experiment No. 4
Aim: Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.
Name: Saniya Laxman Parab
Roll Number: 52
Date of Performance:
Date of Submission:

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	20	

Checked by

Name of Faculty :

Signature :

Date



Experiment 4

Title

Develop a dynamic webpage where content updates on button click using JavaScript DOM. Implement jQuery effects: hide/show elements, toggle visibility, and animate elements.

Theory:

1. JavaScript and the DOM (Document Object Model):

- The **DOM** is a hierarchical representation of the elements on a webpage. It allows programming languages like JavaScript to interact with the structure, style, and content of the page.
- Using **DOM methods**, we can:
 - Access elements (getElementById, getElementsByClassName, querySelector)
 - Change content (innerHTML, textContent)
 - Modify attributes (setAttribute, removeAttribute)
 - Change styles dynamically (style.property)
- Example: Changing the text of a paragraph when a button is clicked enhances user interactivity.

2. Dynamic Webpage Content:

- A **dynamic webpage** changes content without reloading the page.
- Dynamic updates improve **user experience** by allowing real-time feedback and interactivity.
- In this lab, content updates are achieved using `document.getElementById().innerHTML` to manipulate text dynamically.
-



3. Introduction to jQuery:

- **jQuery** is a lightweight, fast JavaScript library that simplifies:
 - HTML document traversal and manipulation
 - Event handling
 - Animations and effects
- jQuery is **cross-browser compatible**, meaning the same code works in different browsers without modification.

Syntax

Example:

`$(selector).action();`

- - `$()` is the jQuery function
 - `selector` identifies the HTML element
 - `action()` is the method applied (like `hide()`, `show()`, `animate()`)

4. jQuery Effects and Animations:

- **Hide/Show:** Instantly hide or show elements.
- **Toggle:** Switch visibility state of an element (hide → show, show → hide).
- **Animate:** Allows smooth transitions of CSS properties (like height, width, opacity, margins).
- **Benefits of jQuery Effects:**
 - Improves visual interactivity
 - Reduces manual CSS manipulation
 - Provides smoother user experience



5. Integration of JS DOM and jQuery:

- Using **pure JavaScript** allows control over element content and structure.
- Using **jQuery** complements JS by handling animations and effects more efficiently.
- Combining both allows for **dynamic and interactive web pages** where content changes and user interactions are smooth and visually appealing.

6. Applications:

- Dynamic webpages are used in:
 - Online forms that validate inputs instantly
 - Content sliders and carousels
 - Real-time notifications
 - Interactive dashboards

Procedure

1. Include jQuery CDN:

```
<script src="https://code.jquery.com/jquery-3.7.0.min.js"></script>
```

2. Create HTML element:

```
<div id="box">Hello</div>  
<button id="btn">Toggle</button>
```

3. Apply jQuery:

```
<script>  
$('#btn').click(function(){  
    $('#box').toggle(500);  
});  
</script>
```



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CODE:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <title>Dynamic Webpage with DOM & jQuery</title>
```

```
  <script src="https://code.jquery.com/jquery-3.7.1.min.js"></script>
```

```
<style>
```

```
  body {
```

```
    font-family: Arial, sans-serif;
```

```
    text-align: center;
```

```
    margin-top: 40px;
```

```
  }
```

```
  #box {
```

```
    width: 120px;
```

```
    height: 120px;
```

```
    background-color: steelblue;
```

```
    margin: 30px auto;
```



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```
position: relative;
}
```

```
button {
    padding: 10px 15px;
    margin: 5px;
    cursor: pointer;
    background-color: aquamarine;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2 id="text">Click the button to change this text</h2>
```

```
<button onclick="changeText()">Change Text (DOM)</button>
```

```
<br><br>
```

```
<button id="hideBtn">Hide Box</button>
```

```
<button id="showBtn">Show Box</button>
```



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```
<button id="toggleBtn">Toggle Box</button>
```

```
<button id="animateBtn">Animate Box</button>
```

```
<div id="box"></div>
```

```
<script>
```

```
function changeText() {  
    document.getElementById("text").innerHTML = "Text updated  
using JavaScript DOM!";  
}  
$(document).ready(function ()  
{  
    $("#hideBtn").click(function ()  
    {  
        $("#box").hide(800);  
    });  
  
    $("#showBtn").click(function () {  
        $("#box").show(800);  
    });  
});
```



```
$("#toggleBtn").click(function () {  
    $("#box").toggle(800);  
});  
  
$("#animateBtn").click(function () {  
    $("#box").animate({  
        left: "150px",  
        width: "150px",  
        height: "150px",  
        opacity: 0.5  
    }, 1000).animate({  
        left: "0px",  
        opacity: 1  
    }, 1000);  
});  
  
});  
</script>  
</body>  
</html>
```




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OUTPUT

Click the button to change this text

Change Text (DOM)

Hide Box

Show Box

Toggle Box

Animate Box

Text updated using JavaScript DOM!

Change Text (DOM)

Hide Box

Show Box

Toggle Box

Animate Box





Conclusion

JavaScript DOM allows dynamic updating of webpage content in real-time. jQuery simplifies element manipulation, effects, and animations. Combining both creates interactive and visually appealing webpages, enhancing user experience. This lab demonstrated practical implementation of content updates, visibility toggling, and animations.

Question:

1. What is the difference between .hide() and .toggle() in jQuery?

.hide() hides the element.

.toggle() switches the element between show and hide each time it is called.

2. How can you stop an ongoing animation in jQuery before it completes?

You can use the **.stop()** method to stop the current animation immediately.

Example: `$("#box").stop();`

3. How can you use jQuery to fade in and fade out an element?

Use **.fadeIn()** to show the element slowly and **.fadeOut()** to hide it slowly.

Example:

`$("#box").fadeIn();`

`$("#box").fadeOut();`